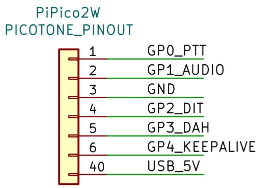
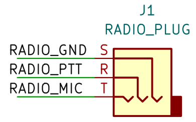
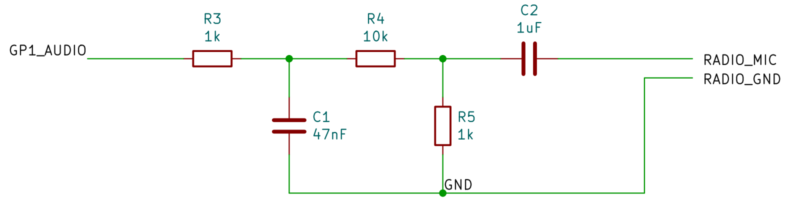
PICO INTERFACE



RADIO INTERFACE

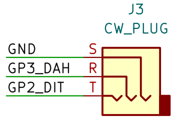


AUDIO INTERFACE

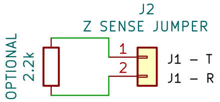


R3/C1 soften the GPIO square wave. R4/R5 attenuate roughly 11:1. C2 AC–couples to the radio mic input.
These values are a safe starting point, not a radio–specific final calibration.

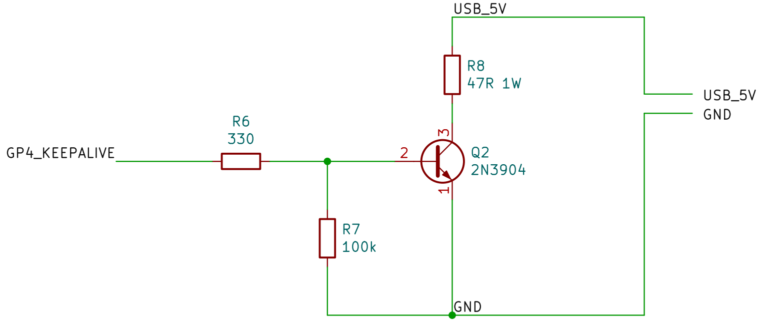
KEY INTERFACE



IMPEDANCE SENSED PTT

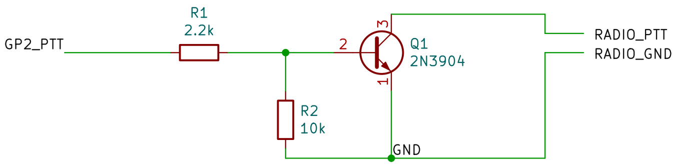


USB POWER–BANK KEEP–ALIVE LOAD



Q2 is a saturated low–side switch for the USB power–bank dummy load.
Firmware pulses GP18 for 1 second every 5 seconds by default.
R8 dissipates about 0.53 W during each pulse at 5 V, so use at least a 1 W resistor.

PTT INTERFACE



Q1 provides open–collector style PTT: GP15 HIGH turns Q1 on and shorts RADIO_PTT to GND.

DESIGN NOTES

1. Pico and radio grounds are common in this implementation.
2. The PTT circuit assumes the radio keys by shorting PTT to ground. Verify the target radio connector before building.
3. IMPEDANCE–SENSED PTT: For radios such as the Yaesu FT–65R, install a 2.2 kΩ resistor across J2 pins 1 and 2. Connect the radio's shared MIC/PTT conductor ONLY to J1 – T. Do not connect the radio directly to J1 – R. Do not leave R of the radio connected. Connect radio ground to J1 – S.

Keep–alive load: 47 ohm 1W gives about 106 mA / 0.53 W at 5 V
Audio attenuation values are a starting point; adjust for radio mic level
PTT transistor pulls radio PTT to ground when GP15 is high
GP15 PTT, GP16 audio, GP18 USB power–bank keep–alive

PicoTone

Sheet: /		
File: picofox_interface.kicad_sch		
Title: PicoTone Interface – AF0FR		
Size: A4	Date: 2026–09–01	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1